

Linear regression and causality

ECON726

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Regression and causality

Regression gives the best linear approximation to the CEF, in a MMSE sense

- ▶ But when does a regression represent something causal?

A regression is **causal** when the CEF it approximates is causal

- ▶ Helpful to think of a CEF being causal by using the potential outcomes framework
- ▶ CEF is causal when it describes differences in average potential outcomes for a fixed reference population

Review - Potential Outcomes framework

Consider the relationship between future earnings Y_i and whether i attends college, denoted by D_i

$D_i \in \{0, 1\}$	Binary treatment variable for individual i
Y_i	Measure of earnings for individual i
Y_{1i}	Earnings if individual i goes to college ($D_i = 1$) Alternative notation: $Y_i(1)$, Y_i^1 , y_{i1}
Y_{0i}	Earnings if individual i does not go to college ($D_i = 0$) Alternative notation: $Y_i(0)$, Y_i^0 , y_{i0}

- ▶ Y_{0i} is earnings of an individual had they not gone to college, *irrespective of whether they actually went*
- ▶ Causal effect college attendance for individual i is the difference $Y_{1i} - Y_{0i}$

Observed outcome Y_i can be written as:

$$\begin{aligned} Y_i &= \begin{cases} Y_{1i} & \text{if } D_i = 1 \\ Y_{0i} & \text{if } D_i = 0 \end{cases} \\ &= Y_{0i} + (Y_{1i} - Y_{0i})D_i \end{aligned}$$

We can see Y_{0i} or Y_{1i} in the data, but never both

Review - Fundamental problem of causal inference

Problem:

- ▶ If an individual is treated ($D_i = 1$), we never observe the outcome in which they are not treated (Y_{i0})
- ▶ If an individual is not treated ($D_i = 0$), we never observe the outcome in which they are treated (Y_{1i})

The causal effect $Y_1 - Y_0$ is unidentifiable

Group	Y_1	Y_0
$D = 1$	Observable as Y	Unobservable counterfactual
$D = 0$	Unobservable counterfactual	Observable as Y

Y_1	Y_0
y_{11}	y_{10}
y_{21}	y_{20}
y_{31}	y_{30}
$\vdots$	$\vdots$
y_{n1}	y_{n0}

Naïve comparison: $\mathbf{E}[Y|D = 1] - \mathbf{E}[Y|D = 0]$

Let's consider the averages of Y_{1i} , Y_{0i} , and Y_i :

$$\begin{aligned}\mathbf{E}[Y|D = 1] &= \mathbf{E}[Y_0 + (Y_1 - Y_0)D|D = 1] && \because Y = Y_0 + (Y_1 - Y_0)D \\ &= \mathbf{E}[Y_1|D = 1] \\ \mathbf{E}[Y|D = 0] &= \mathbf{E}[Y_0|D = 0]\end{aligned}$$

We can re-write the observed difference in health outcomes as:

$$\mathbf{E}[Y|D = 1] - \mathbf{E}[Y|D = 0]$$

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Naïve comparison (cont.)

Observed difference in earnings equals the average treatment effect (ATE) on the treated plus selection bias:

$$\mathbf{E}[Y|D = 1] - \mathbf{E}[Y|D = 0] = \underbrace{\mathbf{E}[Y_1 - Y_0|D = 1]}_{\text{ATE on treated}} + \underbrace{\mathbf{E}[Y_0|D = 1] - \mathbf{E}[Y_0|D = 0]}_{\text{Selection bias}}$$

ATE on treated is the average causal effect of college attendance on those who went to college:

$$\mathbf{E}[Y_1|D = 1] - \mathbf{E}[Y_0|D = 1] = \mathbf{E}[Y_1 - Y_0|D = 1]$$

Selection bias:

$$\mathbf{E}[Y_0|D = 1] - \mathbf{E}[Y_0|D = 0]$$

What do you think selection bias looks like in this example?

Naïve comparison (cont.)

Observed difference in earnings equals the average treatment effect (ATE) on the treated plus selection bias:

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$$\mathbf{E}[Y_1|D = 1] - \mathbf{E}[Y_0|D = 1] = \mathbf{E}[Y_1 - Y_0|D = 1]$$

Selection bias:

$$\mathbf{E}[Y_0|D = 1] - \mathbf{E}[Y_0|D = 0]$$

What do you think selection bias looks like in this example?

- ▶ The difference in Y_{0i} between those who went to college and those who didn't
- ▶ Consider someone who went to college. Now assume that person didn't go. Their expected earnings may be higher than those of someone who didn't go to college at all
 - ⇒ Those who didn't go to college, on average, have lower values of Y_{0i}
 - ⇒ Selection bias is positive

Random assignment and selection bias

Random assignment ($Y_0, Y_1 \perp D$) will solve the selection problem

- Causes selection bias to equal zero

$$\mathbf{E}[Y_1|D = 1] = \mathbf{E}[Y_1|D = 0] = \mathbf{E}[Y_1] \quad \text{and} \quad \mathbf{E}[Y_0|D = 0] = \mathbf{E}[Y_0|D = 1] = \mathbf{E}[Y_0]$$

$$\implies \mathbf{E}[Y_0|D = 1] - \mathbf{E}[Y_0|D = 0] = 0$$

$\implies$ Selection bias is 0

$\implies$ Naïve comparison = ATE on treated

What do we do when we don't have random assignment?

Conditional Independence Assumption

Independence of potential outcomes and the treatment:

$$\Rightarrow (Y_0, Y_1 \perp D)$$

$\Rightarrow$ randomization

What happens when we don't have independence? Meaning the occurrence of D influences the occurrence of Y_0 and Y_1 (or vice versa)?

Perhaps we can condition on certain variables X , such that the treatment D_i and the potential outcomes Y_{1i}, Y_{0i} LOOK independent

$$\{Y_{0i}, Y_{1i}\} \perp D_i | X_i$$

Conditional independence

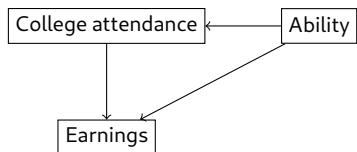
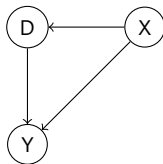
Conditional independence assumption:

$$\{Y_{0i}, Y_{1i}\} \perp D_i | X_i$$

Intuitively: occurrence of D_i does not influence the occurrence of Y_{0i} or Y_{1i} , or vice versa, conditional on knowing X_i :

$$\mathbf{E}[Y_{0i} | D_i, X_i] = \mathbf{E}[Y_{0i} | X_i]$$

$$\mathbf{E}[Y_{1i} | D_i, X_i] = \mathbf{E}[Y_{1i} | X_i]$$



Selection bias under CIA

Under conditional independence assumption, selection bias disappears conditional on X_i :

$$\begin{aligned}\mathbf{E}[Y_{0i}|X_i, D_i = 1] - \mathbf{E}[Y_{0i}|X_i, D_i = 0] &= \mathbf{E}[Y_{0i}|X_i, D_i = 0] - \mathbf{E}[Y_{0i}|X_i, D_i = 0] \\ &= 0\end{aligned}$$

So we can write the difference in observed outcomes **conditional on X_i** as:

$$\begin{aligned}\mathbf{E}[Y_i|X_i, D_i = 1] - \mathbf{E}[Y_i|X_i, D_i = 0] &= \mathbf{E}[Y_{1i}|X_i, D_i = 1] - \mathbf{E}[Y_{0i}|X_i, D_i = 0] \\ &= \mathbf{E}[Y_{1i}|X_i] - \mathbf{E}[Y_{0i}|X_i] \\ &= \mathbf{E}[Y_{1i} - Y_{0i}|X_i]\end{aligned}$$

Given the CIA, comparisons of average earnings (conditional on X_i) have a causal interpretation

When treatment takes on more than two values

Expanding CIA assumption to relations that can take on more than two values, such as when $s_i = \{0, 1, \dots, 16\}$ equals years of schooling:

$$Y_{si} = f_i(s)$$

Individual specific function:

- ▶ Potential earnings that person i would receive after obtaining s years of education
- ▶ What would i learn for any value of schooling

If s could only take on two values, $s = 12$ and $s = 16$, then we're considering the value of college vs. no college:

$$Y_{0i} = f_i(12), \quad Y_{1i} = f_i(16)$$

CIA with non-binary treatment

CIA in this more general set-up:

$$Y_{si} \perp s_i | X_i, \quad \forall s_i$$

- ▶ In randomized experiments (RCTs), s_i is randomly assigned based on covariates X_i
- ▶ In observational studies, the CIA means s_i is **as good as randomly assigned** conditional on X_i

Under conditional independence assumption, comparisons of average earnings across schooling levels have a causal interpretation:

$$\mathbf{E}[Y_i | X_i, s_i = s] - \mathbf{E}[Y_i | X_i, s_i = s - 1] = \mathbf{E}[f_i(s) - f_i(s - 1) | X_i]$$

For example, average causal effect of high school graduation:

$$\begin{aligned} \mathbf{E}[Y_i | X_i, s_i = 12] - \mathbf{E}[Y_i | X_i, s_i = 11] &= \mathbf{E}[f_i(12) | X_i, s_i = 12] - \mathbf{E}[f_i(11) | X_i, s_i = 11] \\ &= \mathbf{E}[f_i(12) - f_i(11) | X_i, s_i = 12] \end{aligned}$$

- ▶ Here, selection bias would come from the differences in potential earnings of high school graduates and non-graduates
- ▶ Under CIA, high school graduation is independent of potential earnings conditional on X_i , so selection bias disappears

CIA and regression

Regression is an empirical strategy that turns CIA into causal effects:

- ▶ Assume $f_i(s)$ is linear in s , and the same for everyone, except for an additive term η_i :

$$f_i(s) = \alpha + \rho s + \eta_i$$

- ▶ **Linear constant effects model:** equation is linear, and the functional relationship is the same for everyone
- ▶ Only individual-specific part of $f_i(s)$ is mean-zero η_i , which captures unobserved factors that determine potential earnings

Observed value will be:

$$Y_i = \alpha + \rho s_i + \eta_i,$$

Note that s_i is correlated with η_i : $\mathbf{E}[\eta_i | s_i] \neq 0$. If we ran this regression, we'd expect selection bias to impact our results.

CIA and regression (cont.)

Assume CIA holds, and η_i has the form:

$$\eta_i = X_i' \gamma + v_i$$

where:

- ▶ γ is a vector of population regression coefficients that satisfies $\mathbf{E}[\eta_i | X_i] = X_i' \gamma$
- ▶ v_i and X_i are uncorrelated by construction

Then, because of the CIA:

$$\begin{aligned} \mathbf{E}[f_i(s) | X_i, s_i] &= \mathbf{E}[f_i(s) | X_i] \\ &= \alpha + \rho s + \mathbf{E}[\eta_i | X] \\ &= \alpha + \rho s + X_i' \gamma \end{aligned}$$

The residual in the linear causal model v_i is uncorrelated with s_i and X_i , and ρ is the causal effect of interest:

$$Y_i = \alpha + \rho s_i + X_i' \gamma + v_i$$